PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: Annealing of 1 cm^2 GaAs Chip at 400°C and 450°C in RTA

Requestor: Hannah Alpert, Seungbin Jeong, Anand Lalwani **Date: 12/09/2019**

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Send completed form to: snf-promcommittee@lists.stanford.edu

We are requesting permission to anneal a ~10 x 10 mm chip of AlGaAs/GaAs, partially covered with an Au-Ge-Ni metal stack (43-21-16 nm), in the Allwin 610 RTA. Four temperature/time combinations will be tested: 400°C and 450°C for 30 and 60 seconds.

This request is nearly identical to that by Lucas Peeters on 02/17/2017. As both Lucas Peeters and Xue Bai (02/25/2014) point to, Woodall *et al*. (Woodall *et al.*, App. Phys. Lett. **38**, 639 (1981)) studied the partial pressure of Arsenic as a function of GaAs annealing temperature. The partial pressure decreases strongly with decreasing temperature, and the lowest temperature they measured was 727°C. Thus, at a worst-case scenario (if the outgassing at 450°C were equal to that at 727°C, which it isn't), the outgassing of As would lead to a concentration of 1.7E-12 moles/liter, which is equivalent to 1.7E-9 moles/m^3 or 130 ng/m^3. As Lucas mentioned, this is a factor of 10 times smaller than the recommended exposure level of 2 ug/m^3 and 100 times smaller than the permitted exposure level of 10 um/m^3. However, because the chip only fills a small part of the chamber and the outgassing at 450°C is likely orders of magnitude lower than that at 727°C, the effective concentration will be even lower than the 130 ng/m^3 calculated.

The metal stack poses no health risks and relevant PPE will be worn throughout the process, and an MSDS for a GaAs wafer is attached.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date:

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MATERIAL SAFETY DATA SHEET

I PRODUCT IDENTIFICATION

Trade Name: Gallium Arsenide
Synonym: Gallium Monoarsenide
Formula: GaAs
CAS #: 1303-00-0

II HAZARDOUS INGREDIENTS

Hazardous Components	%	OSHA/PEL	ACGIH/TLV
Gallium Arsenide	0-100	0.5 (As) mg/m^3	0.01 (As) mg/m^3

III PHYSICAL DATA

Boiling Point: N/E
Melting Point: 1238 ^oC
Vapor Density: N/E
Specific Gravity: 5.31 g/cc
Molecular Weight: 144.64
Solubility in H_2O: Insoluble
Appearance and Odor: Dark gray cubic crystals
% Volatiles: N/E or N/A

IV FIRE AND EXPLOSION HAZARDS DATA

Flash Point: N/A
Autoignition Temp: N/E or N/A
Flammable Limits: **Lower**: N/E or N/A **Upper**: N/E or N/A

Extinguishing Media: Use dry chemical, CO_2 or graphite. DO NOT USE WATER OR WATER SPRAY.
Fire Fighting Procedures: Wear NIOSH/MSHA approved self-contained breathing apparatus, flame and chemical resistant clothing, hats, boots, and gloves.
Unusual Fire and Explosion Hazards: Can React with Steam, Heat, Acid and Acid Fumes to Evolve Deadly Poisonous Arsine.

V HEALTH HAZARD INFORMATION

Effects of Exposure:

To the best of our knowledge the chemical, physical and toxicological properties of gallium arsenide have not been thoroughly investigated and reported.

Acute arsenic poisoning from ingestion results in marked irritation of the stomach and intestines with nausea, vomiting and diarrhea. In severe cases, the vomitus and stools are bloody and the patient goes into collapse and shock with weak, rapid pulse, cold sweats, coma and death. Chronic arsenic poisoning may cause disturbances of the digestive system such as loss of appetite, cramps, nausea, constipation or diarrhea.

Gallium and gallium compounds may cause metallic taste, dermatitis and depression of the bone marrow function. Large doses may cause hemorrhagic nephritis.

Acute Effects: May be irritating to skin, eyes and mucous membranes.
Carcinogenicity: **NTP**: Yes **IARC**: Yes **ACGIH**: Yes **OSHA**: Yes

EMERGENCY AND FIRST AID PROCEDURES:

INHALATION: Remove to fresh air; give oxygen if breathing is difficult. Seek medical attention.
INGESTION: Give 2 cups of milk or water and induce vomiting. Seek medical attention.
SKIN: Brush off skin, wash area with soap and water. Seek medical attention.
EYES: Immediately flush eyes, including under eyelids with large amounts of water. Call a physician.

VI REACTIVITY DATA

Stability: Stable
Conditions to Avoid: Water/moisture
Incompatibility (Material to Avoid): Steam, heat, acids, acid fumes.
Hazardous Decomposition Products: Highly toxic fumes of arsenic. Contact with acids releases toxic gases.
Hazardous Polymerization: Will not occur

VII SPILL OR LEAK PROCEDURES

Steps to Be Taken in Case Material Is Released or Spilled: Wear full protective equipment, cover spill with dry sand or vermiculite. Mix well and carefully transfer to a container.

Waste Disposal Method: Dispose of in accordance with Local, State and Federal Waste Disposal Regulations.

VIII SPECIAL PROTECTION INFORMATION

Respiratory Protection (Specify Type): Wear a NIOSH-approved dust-mist-fume cartridge respirator.
Ventilation: Handle under dry protective gas. Handle in a controlled environment. Use local exhaust to maintain below TLV level for Arsenic.

Eye Protection: ANSI approved safety goggles, or glasses.
Protective Gloves: Neoprene, Rubber
Other Protective Equipment: Wear protective clothing to prevent contamination of skin and clothes.

IX SPECIAL PRECAUTIONS

Precautions to Be Taken in Handling and Storage: Handle under dry protective gas. Protect from humidity and water. Store in tightly closed containers in a cool, dry place. Wash hands and face thoroughly after handling and before meals.

Work Practices: Implement engineering and work practice controls to reduce and maintain concentration of exposure. Handle in a controlled, inert atmosphere. Minimize exposure by using local exhaust and an enclosing process if/when possible. Use good housekeeping and sanitation practices. Do not blow dust off clothing or skin with compressed air. Maintain eyewash capable of sustained flushing, safety drench shower and facilities for washing.

The above information is believed to be correct, but does not purport to be all inclusive and shall be used only as a guide. ESPI shall not be held liable for any damage resulting from handling or from contact with the above product.

Issued by: S. Dierks
Date: October 2002